Evolving a Titanium Karambit for Laser Powder-Bed Fusion under Ergonomic, Edge and Serration Constraints

Jepson Taylor

***Abstract*—We report the best individual of the G3 v3.0.0 cohort of record, a one-piece Ti-6Al-4V karambit for laser powder-bed fusion with a hollow, lattice-filled grip and blade core. No genome was set or edited by hand: 24 independent MAP-Elites islands ran 2,500 evaluations each from scrambled Sobol samples, and the design space itself carries the rules a good edge must follow; the decoder, the objective and its rules are human-authored. Both cutting edges have one primary grind of 20.5–20.8° and 20.6–21.6° per side, measured normal to the edge and carried to within 3.0 and 3.1 mm of the point, and both edges are serrated (3 and 5 scallops; the weakest of the teeth between them has a safety factor of 2.83 under a 180 N snag). The design scores 80.55 under W_ERGOLIGHT_v1 draft d with an ergonomic multiplier of 0.988 and its print mesh weighs 59.3 g, which is above the 50 g goal. The 0.4 mm search grid undercounted the mass of the search's own leader (51.3 g against 75.2 g at 0.2 mm and 74.6 g at 0.15 mm), so the ranking of record is measured on the 0.2 mm grid of the print file. Against a safety-factor target of 1, its solid tip (0.64) is below the target and its grip under the spine load (0.74 by voxel FE) is below the target. We describe the representation, objective, search and measurement chain, the evolution that produced the design, and every limit of the result.**

***Index Terms*—Evolutionary design, quality diversity, MAP-Elites, additive manufacturing, Ti-6Al-4V, finite elements, knife design.**

1 Introduction

A karambit is a curved knife with a finger ring at the pommel, held in a closed fist; in this design the little finger goes through the ring. Printing one in Ti-6Al-4V by laser powder-bed fusion (L-PBF) makes a hollow part possible: the grip and the blade core become a skin over a triply periodic minimal-surface (TPMS) lattice with ports to remove the powder. That opens a large three-dimensional design space, and with it several ways for a search to fool itself. A coarse voxel grid can hide mass. A beam or plate formula can credit material that is never built. A blade can keep an unground shoulder where a sparse sharpness check does not look. And deep, narrow serrations can lengthen the edge while leaving needle teeth that snap on the first hard snag.

G3 is the generation of the Jarambit project that designs the whole knife around the hand: the ring, the grip, the guard, the hollow interior and a double-edged, double-serrated claw blade come from one decoder and are scored by one multi-fidelity evaluation chain. This paper reports its v4.0.1 release and the best individual of the cohort of record, `hz4i013_g00995_d206e3c9` (decoder commit `80e669e5b558`, decoder hash `a203278d80d2`).

Three choices define the release.

1) *Evolution designs the knife.* Every island starts from scrambled Sobol samples (g3_genome.rs); no seed design, template or hand edit enters a population. Engineering requirements enter either as constraints on what the decoder can express or as terms of the objective.
2) *The edge is part of the design space, not a post-process.* The decoder gives both cutting edges one primary grind at 17.5–22.5° per side, measured normal to the actual serrated edge and run unbroken to the point (EDGE-1..5), and serrates both edges within bounds chosen so that the teeth do not snap (SERR-1..4; Section 2). The strength of every tooth between two scallops is checked under a 180 N snag and a 60 N pry, on 60 Sobol designs in the decoder's tests and on the released design (serration_to oth_strength).
3) *Every computed number has one source.* Each computed quantity of the design in this paper and in the repository README is generated from one release receipt, evid ence/g3/v3.0.0/release.json, by g3-release-tex and ge n_readme.py; the rule constants (gene bounds, loads, allowables) are the decoder's own and are quoted as written. The measurements are voxel geometry, plate and solid finite elements and a voxel finite-element check of the printed part; three quantities are not finite-element results and are labelled where they appear: the buckling factor (an analytic estimate), the tooth safety factors (a beam formula on the exact section law) and the edge geometry (measured on the exact section law rather than on voxels).

Section 2 describes the genome and the edge and serration rules, Section 3 the objective, Section 4 the search and its evaluation tiers, Section 5 the measurement and verification chain, Section 6 the evolution and the design it found, Section 7 the limits of the result and Section 8 how to reproduce it.

2 Design space

2.1 Genome and decoder

A G3.0 genome is a vector of 179 genes $\mathbf{u} \in [0,1]^{179}$ (g3_genome.rs). G3Genome::decode maps it to named physical parameters, from which the decoder builds a signed distance field (SDF) of the whole knife (g3_sdf.rs).

Release v4.0.1, decoder commit `80e669e5b558` (decoder hash `a203278d80d2`). Every computed number and every figure of the design in this paper is generated from the release receipt evidence/g3/v3.0.0/release.json. Status: computational design; not built, printed or physically tested.

- **Ring and grip.** The finger ring has a fixed 24.9 mm bore and a fixed 8 mm lip width; the lip's radial depth is a gene (1.5–2.8 mm). A filleted junction joins it to a compact grip with a palm swell, a variable section and three finger grooves (ring, middle and index finger). The grip surface carries a procedural texture blended from five bases: knurl, a sinusoidal stripe pattern, Voronoi cells, longitudinal ridges and a small compositional pattern-producing network.
- **Guard and blade.** An integral guard leads to a curved claw blade whose straight-line reach from the guard is capped at 85.725 mm (3.375 in, BLADE_REACH_MAX_MM); its hook drop is a gene. The blade thickness tapers linearly from a core value (4–7 mm) to a tip value (1.5–4.5 mm), with a dorsal ridge and fuller, and the last 6–12 mm before the point follow a five-knot monotone profile.
- **Hollow interior.** Inside a skin of 0.8–1.2 mm, the grip, guard and blade core are filled with a blend of Gyroid and Schwarz-D lattices (tpms_val, tpms_level_for_rho) with stiffening ribs and shear webs, and powder-removal ports open the cavity to the outside. The hand-zone mask is cleaned by variable-radius morphological closing and opening, and enforce_am_connectivity makes the build printable: air is flooded from the outside of the grid, any sealed pocket is filled solid (so trapped powder is never counted as saved mass), and only the largest connected solid is kept.

The decoder revision of this release, r2, is identified by the SHA-256 hash `a203278d80d2`, computed by g3_decoder_sha256 (g3_qd.rs) over the decoder's source files, the shared physics and CAD sources they call, and both lock files; an archive written under another hash fails its integrity check.

2.2 Edge rules (EDGE-1..5)

Both the inner edge (edge A) and the spine edge (edge B) are cutting edges, and five rules make each a real grind rather than a chamfer on a strut.

1) **EDGE-1, one full-height grind.** Each edge has a single primary grind, carried outward until it meets the blade body, with no micro-bevel and no shoulder; at the guard-to-blade seam the grind takes precedence over the hand-zone rounding, so the plunge keeps its thin bevel.
2) **EDGE-2, normal to the real edge.** The bevel depth is the exact distance d to the actual, serrated edge polyline, so the angle is the same around the curved claw and inside every scallop.
3) **EDGE-3, the angle.** The per-side angle θ is a gene bounded to 17.5–22.5° (35–45° inclusive, centred on 20°, a hard-use edge for titanium), and the apex flat t_{apex} is 0.05–0.12 mm. The grind thickness is

$$t(d) = t_{\text{apex}} + 2\,d\tan\theta\,(1 + 0.18\,c\,\min(1, (d - 0.4)^{+}/6)), \tag{1}$$

   where the grind-curvature gene $c \in [-0.35, 0.45]$ makes the grind slightly hollow or convex beyond the first 0.4 mm, by −6 to +8 % in slope. This is why the bevel angles in Table 5 span a little more than the apex angles.
4) **EDGE-4, to the point.** Both grinds run unbroken to the point. Only the plunge by the guard is unsharpened: the grind begins 5–12 % of the way along the edge (a gene) and ramps in over 3 mm. Each sharpened edge is capped at 101.6 mm (4.0 in).
5) **EDGE-5, the check.** The sharpness gate (measure_blade_edge_apex_and_angle) samples the exact section law along the inward edge normal at 0, 0.4 and 2.0 mm behind the apex, at every station whose grind is steady, on both edges. The worst station sets the sharpness terms, so a blunt station or a shoulder behind the apex costs score.

2.3 Serration rules (SERR-1..4)

Both edges are serrated, under four rules.

1) **SERR-1, depth.** The serration depth is 0.8–2.2 mm on the inner edge and 0.8–2.0 mm on the spine edge, so no design has a plain edge.
2) **SERR-2, count.** The pitch is capped on the part of the span cut to full depth, so every serrated span holds at least three full-depth scallops.
3) **SERR-3, placement.** The span starts 16–24 % of the way along the blade, past the plunge, and ends at least $\max(\text{tip reserve} + 3, 18)$ mm before the point, so the piercing point stays plain.
4) **SERR-4, teeth that do not snap.** The pitch is 5–9 mm and the depth at most a quarter of the pitch.

These bounds are what the search sees. The strength of the resulting teeth is checked separately (serration_tooth_strength), on 60 Sobol designs in the decoder's tests and on the released design; it is not a term of the score. Every tooth between two neighbouring scallops is checked on the exact 2D section law. Its root is the segment of length b between the two gullet bottoms, and its height h_t is the outline's furthest point from that segment. Sampling the blade thickness $t(s)$ at 40 points along the root gives the in-plane second moment I_{ip} about the root's thickness-weighted centroid, the extreme-fibre distance c from that centroid, the out-of-plane second moment $I_{\text{op}} = \int t^3/12\,\mathrm{d}s$ and the largest thickness $t_{\max}$. Two load cases act at the tooth tip, a snag of $F_{\text{s}} = 180$ N along the edge and a pry of $F_{\text{p}} = 60$ N across the blade:

$$\sigma_{\text{snag}} = K_t\,\frac{F_{\text{s}}\,h_t\,c}{I_{\text{ip}}}, \qquad K_t = 1 + 2\sqrt{h_t/\rho}, \tag{2}$$

$$\sigma_{\text{pry}} = \frac{F_{\text{p}}\,h_t\,t_{\max}/2}{I_{\text{op}}}, \tag{3}$$

where ρ is the smaller radius of the circles through each gullet bottom and the outline ± 0.6 mm either side of it (the Inglis notch factor [7]). The tooth's safety factor is $755\ \text{MPa}/\max(\sigma_{\text{snag}}, \sigma_{\text{pry}})$, using the Ti-6Al-4V yield stress of 828 MPa derated by 0.912. The decoder's tests require at least 1 for every checked tooth of the 60 designs.

3 Objective

Designs are ranked by W_ERGOLIGHT_v1, draft d, under the requirement set RS-karambit-v3 (ergolight.rs):

$$\text{score} = 100\,E\sum_{i=1}^{21} w_i\,s_i, \qquad E = \prod_{k=1}^{8} e_k^{\beta_k}, \tag{4}$$

Table 1 | Weights w_i and ergonomic exponents β_k of W_ERGOLIGHT_v1 (draft d)

Term s_i	w_i	Factor e_k	β_k
Mass	0.3700	Ring lip	0.18
Solid tip (TET10)	0.0700	Grip section	0.18
Gate-S edge sharpness	0.0500	Hand-edge radii	0.18
Gate-S point profile	0.0500	Finger index	0.15
AM build	0.0500	Balance	0.10
CAD watertight	0.0400	Grip length	0.08
Karambit ring and claw	0.0400	Guard height	0.08
Form fairness	0.0400	Grip surface	0.05
Lateral, 180 N	0.0305		
Lateral, 260 N	0.0305		
Yield sequence (≥ 1.2)	0.0305		
Ring back (30 N, SF 2)	0.0261		
Ring bearing (300 N, SF 2)	0.0218		
Blade-body twist	0.0218		
Blade-tip twist	0.0218		
Plate GCI (3 levels)	0.0200		
Strict lateral, 260 N	0.0174		
Buckling λ	0.0174		
Human-max twist and spine	0.0174		
Fatigue (Goodman)	0.0174		
Hollow-grip strength	0.0174		

where the 21 term scores $s_i \in [0, 1]$ carry weights $\sum_i w_i = 1$, and $E \in [0.3, 1]$ is a geometric-mean ergonomic multiplier over eight hand-zone factors $e_k \in [0.3, 1]$ with $\sum_k \beta_k = 1$ (Table 1). Because E multiplies the whole sum instead of adding a bonus, a hand-zone shortfall scales down every point earned elsewhere. It does not forbid trading comfort for grams: at the best individual, 1 % of E is worth 2.3 g of mass credit (0.348 points per gram). Once all eight measurements are inside their target bands, $E = 1$, and ergonomics no longer distorts the trade between mass and strength.

Mass carries the largest weight, $w = 0.370$, mapped linearly from 0 at 130 g to 1 at 25 g. In draft d the solid-tip term carries 0.070, to push the point to resist the 1.82 N·m twist, and the two Gate-S sharpness terms and buildability carry 0.050 each. Every safety-factor ratio $v = \mathrm{SF}/\mathrm{SF}_{\mathrm{req}}$ passes through a knee function (sfk),

$$\mathrm{sfk}(v) = 0.94\ \min(\max(v, 0), 1) + 0.06\,(1 - e^{-\max(0, v-1)/0.15}), \tag{5}$$

which gives 94 % of a term's credit linearly up to adequacy ($v = 1$) and the last 6 % for margin above it, so reserve is rewarded without paying for mass that adds nothing. No term is a hard gate: a design below a target loses score in proportion, and the search can still learn from it.

Three terms are fixed or capped in this release, and the scores in this paper include them as they are.

- The plate-GCI term is held at 0.5, because no three-level plate refinement runs in the search.
- The solid-tip term is capped at 0.8 of its knee value, because the independent cross-check that would lift the cap is not run.
- The buildability term is the analytic rule (skin at least 0.8 mm, open ports, lattice struts against a 0.8 mm minimum) at every tier.

4 Search

4.1 Archive, start and variation

Each island is a MAP-Elites archive [1], [2] of 1,024 niches in an eight-dimensional behaviour space (CvtMapEl itesArchive). The niche centres are the first 1,024 points of a Halton sequence, so every island shares the same partition. The descriptors are hook drop, inner turn angle, silhouette complexity, grip section height-to-width ratio, finger-groove articulation, porosity, surface-texture energy and mass. The last three are re-measured on the voxel model at Tier 1 and above; the other five come from the genes and the 2D profile.

An island starts from 64 Joe–Kuo Sobol points in $[0, 1]^{179}$ [3], scrambled with a random linear matrix and a digital shift [4], and from nothing else. Afterwards each evaluation breeds one child from two archived parents. With probability 0.4 both parents are drawn from the 16 best-scoring elites; otherwise each parent wins a tournament of three on curiosity, a niche score that rises when the niche's offspring enter the archive and decays when they do not. reproduce_iso_line_dd then combines three steps:

- with probability 0.15, a crossover that swaps one whole gene module from the second parent;
- self-adaptive step sizes, one per gene module, carried in five genes of the genome (0.01–0.20) and mutated log-normally;
- an iso-line step: Gaussian noise at the module's step size plus $0.25\,\mathcal{N}(0, 1)$ times the difference to the second parent.

A child takes its nearest niche if the niche is empty, if the child was evaluated at a higher tier than the incumbent, or if it scores higher at the same tier (CvtMapElitesArchive::insert).

4.2 Three-tier evaluation

To spend finite-element time on promising designs, every child passes through a three-tier cascade.

1) **Tier 0, analytic.** Closed-form geometry and beam screens on the decoded 2D profiles, including a lateral-torsional buckling estimate, take a few milliseconds. A design scoring at least 62 goes on to Tier 1.
2) **Tier 1, voxel model and plate.** A voxel SDF of the whole knife and its lattice at $h = 0.4$ mm (build_g3 _sdf) gives mass, centre of mass, spin inertia and the grip section. One sandwich-plate finite-element mesh of the blade (CST membrane and DKT bending) gives the lateral capacity under a unit load, the blade twist at 1.82 N·m (scaled linearly to the 16.2 N·m human maximum) and the stress under the 3,427 N spine load. A T6 membrane model of the ring band gives its bearing and back capacities. A Tier-1 score of at least 71 promotes the design to Tier 2.
3) **Tier 2, mesh and solid point.** A watertight marching-tetrahedra mesh of the voxel field is censused: no boundary edges and no non-manifold edges. A solid quadratic-tetrahedron (TET10) model of the last 30 mm of the blade is then solved under a 1.82 N·m twist about the axis from the coordinate origin to the point, on a planform mesh of at most 0.15 mm^2 triangles (evaluate_tier2_elites).

Two analytic quantities are carried unchanged into every tier's record, and Table 5 labels both. The buckling factor is an analytic estimate at every tier. The Gate-S edge geometry is measured on the exact section law rather than on voxels. Everything else reported from the search is a Tier-1 or Tier-2 measurement.

4.3 Islands and logging

The islands are independent; each is seeded from its host and island numbers (g3-evolve). Every 50 evaluations (--log-every 50) an island appends a G3HistoryPoint to history.jsonl. Each point records the evaluation count, the filled niches, the Tier-2 count, the QD score, and the score, mass and tier of the best elite (ranked by tier first, then score). At power-of-two evaluation counts and at the end, the point also records that elite's full genome row, which Section 6 renders. Logging draws no random numbers, so a logged run produces the same archive as an unlogged one; a test checks this.

5 Measurement and verification

5.1 Platform and grid

The cohort ran on xbabe4 (arm64 macOS). Floating-point libraries differ between platforms, so the final ranking was repeated on Linux x86_64 (xbabe2), Tier 2 at h 0.2 mm, which is the platform of record for every number below; the winner scored 80.55 on the search host and 80.55 on the platform of record. The search grid is h = 0.4 mm and the ranking grid h = 0.2 mm (Section 6). Mass is reported from the watertight 0.2 mm print mesh (13,385 mm^3 at 0.00443 g/mm^3, 59.3 g); the ranking counted 57.0 g on its voxel grid and the search 56.0 g.

5.2 Solid tip

The point is checked by a solid TET10 model of the last 30 mm of the blade under a 1.82 N·m twist about the axis from the coordinate origin to the point, reading the peak nodal von Mises stress 4–15 mm from the point. Three planform meshes give safety factors of 0.585, 0.638 and 0.611 at maximum triangle areas of 0.4, 0.15 (production) and 0.08 mm^2. The series is not monotone, so it shows the mesh sensitivity rather than a converged value, and the thickness raster that feeds the model is offset by half a cell (Section 6.4).

5.3 Voxel finite-element check

The plate and beam models of the search are fast surrogates. For the winner we also solved the printed geometry itself: a hex8 linear-elastic model on its AM-build voxels at h = 0.4 mm (skin, lattice, ribs and ports as built), in two zones, the grip and the blade root. Each zone is clamped at one end and loaded at the other with the wrench of three cases: 260 N lateral at the tip, the 16.2 N·m human-maximum twist, and 3427 N on the spine at the blade centroid. We report the 99.9th-percentile von Mises stress of each zone against an allowable of 674 MPa, not a single-element peak, because the voxel staircase concentrates stress artificially at re-entrant corners. Table 2 compares the two.

Table 2 | Safety factors of the winner: search surrogates and voxel finite elements of the printed geometry

Load case	Grip formula	Grip, voxel FE	Root, voxel FE
Lateral 260 N	5.07	3.55	2.08
Twist 16.2 N·m	12.69	5.50	1.69
Spine 3427 N	0.90	0.74	1.29

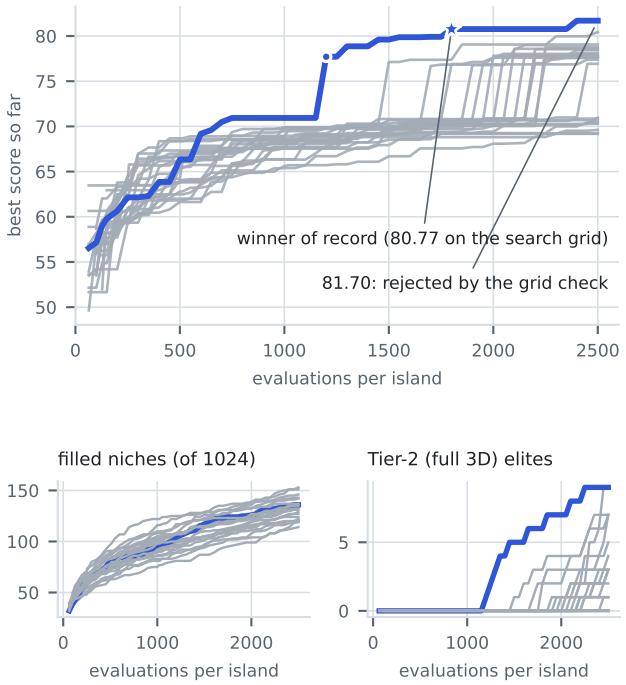


Figure 1 | Top: best score so far of each island (the winning island in colour; the dot marks its first full-3D best). Bottom: filled niches and Tier-2 elites per island.

6 Results

6.1 The evolution

The cohort ran 24 islands of 2,500 evaluations each, 60,000 in all, from 2026-10-26T11:03:49Z to 2026-10-26T12:32:55Z UTC, and kept 3,177 elites, 68 of them at Tier 2. Fig. 1 shows the best score of every island against evaluations, and how the archives filled. The winner came from island13, which started from a best score of 56.54 after its Sobol initialisation. Fig. 2 shows that island's best design at each logged checkpoint, from the same camera.

6.2 Choosing the winner on the print grid

The search measures every design on an h = 0.4 mm grid, and that grid can undercount mass. The search's own leader, `hz4i013_g00840_2a44149e` (Tier-2 score 81.53 at h = 0.4 mm on the platform of record; the search host logged it a little higher, Fig. 2), weighs 51.3 g at h = 0.4 mm but 75.2 g at h = 0.2 mm and 74.6 g at h = 0.15 mm, and its Tier-1 score falls from 75.06 to 66.65 (Table 3). The two finest grids agree within 3.3 % for every design, so the ranking and the mass of record are measured at h = 0.2 mm, the grid of the print file; the voxel finite-element check (Section 5) runs at h = 0.4 mm.



eval 64	eval 128	eval 256	eval 512	eval 1,024	eval 2,048	eval 2,500
56.54, 72.1 g	59.01, 65.3 g	62.15, 68.0 g	66.34, 59.9 g	70.94, 62.4 g	80.77, 56.0 g	81.70, 51.3 g
					winner of record	**rejected by the grid check**

Figure 2 | Best design of island13 at each logged checkpoint (a checkpoint sequence, not a parent–child lineage): evaluation, best score and mass on the search grid. Same camera, light and material for every frame; geometry is the exported mesh.

Table 3 | Grid check: mass (g) of the top 8 of the search-grid ranking at five voxel pitches h (mm), Tier 1

Design	0.4	0.3	0.25	0.2	0.15
`hz4i013_g00840_2a44149e`	51.3	55.6	54.9	75.2	74.6
`hz4i013_g00995_d206e3c9`	56.0	60.0	59.0	57.0	57.5
`hz4i002_g00995_6f4643f1`	58.8	59.6	58.9	58.7	58.5
`hz4i013_g00345_be45c4bc`	55.3	59.7	56.4	56.7	57.1
`hz4i002_g00479_76f5e0ac`	58.0	61.7	58.2	59.5	58.9
`hz4i013_g00555_5cd1f106`	50.6	54.1	54.4	52.1	53.1
`hz4i005_g00345_c7922cf1`	56.8	61.0	57.6	58.5	59.7
`hz4i002_g00785_6b2ff4aa`	59.1	62.0	58.8	61.8	59.8

Table 4 | Final ranking: Tier 2 at $h = 0.2$ mm on the platform of record

Rank	Design	Score	Mass (g)
1	`hz4i013_g00995_d206e3c9`	80.55	57.0
2	`hz4i002_g00995_6f4643f1`	80.44	58.7
3	`hz4i013_g00345_be45c4bc`	79.51	56.7
4	`hz4i002_g00479_76f5e0ac`	79.39	59.5
5	`hz4i013_g00555_5cd1f106`	79.27	52.1
6	`hz4i004_g00501_cb3c9e4e`	79.24	57.3
7	`hz4i002_g00905_b274d4d9`	78.94	59.6
8	`hz4i007_g00005_98a9344b`	78.77	53.7

The final ranking ordered every elite by its record from the run, re-evaluated the top 32 at Tier 2 at $h = 0.2$ mm, and the top 8 again on the platform of record (Table 4). The top eight span 80.55–78.77, a spread the measurement chain does not resolve, so *best* below means first of these eight.

6.3 The best individual

The best individual, `hz4i013_g00995_d206e3c9` (genome hash `d206e3c9f4c3`), scores 80.55 with a weighted base of 0.815 and an ergonomic multiplier of 0.988. Its grip is 64.9 mm long with a swell perimeter of 64.4 mm and a height-to-width ratio of 1.59; the guard is 11.4 mm high and the reach of the blade from the guard is 69.3 mm. Its print mesh weighs 59.3 g (Table 6 lists the four masses of this design). Fig. 3 shows it; Table 5 lists its measurements.

6.4 Diagnostics added in v3.0.3

Four diagnostics re-examine this design without changing its receipt or the ranking (g3-diagnostics; evidence/g3/v3.0.0/diagnostics/).

- **Yield sequence per load case.** The ratio in Table 5 divides the grip's worst safety factor, set by the spine load, by the blade's under the 260 N lateral load, so it mixes two load cases. Taken per case with the voxel-FE grip factors, the ratios are 2.74 (lateral), 5.06 (twist) and 0.52 (spine); only the spine case is below 1.2.
- **Fatigue.** The coded fatigue stress divides the plate peak by the thin-wall knock-down 0.891 that the allowable already carries. Applied once, the factor is 1.06 instead of 0.94.
- **Solid tip.** The thickness raster fed to the TET10 model is filled at cell corners and read at cell centres, a half-cell offset. With the offset removed, the production mesh gives 0.55 instead of 0.64. The point is blunt in plan view: its included angle is 110, 99 and 92° at 1, 2 and 3 mm from the point (4.7 mm wide at 2 mm), which the thickness-only point gate does not measure.
- **Plate model.** Refining the search's blade plate mesh to maximum element areas of 3.5, 1.0, 0.25 and 0.0625 mm^2 gives lateral capacities of 337, 191, 254 and 103 N. The series does not converge, as a peak stress at sharp features does not, so the plate's lateral, twist and spine values are values of the search model, not converged estimates.
- **Mass.** Table 6 lists the four masses of the design. Scored at its print-mesh mass it would score 79.75 instead of 80.55. The grip holds 48 % of the printed mass, the blade 33 % and the ring 15 %.

Table 5 | Results for the best individual (platform of record; each model as labelled)

Measure	Value
Mass, print mesh (watertight, 0.2 mm)	59.3 g
Edge A per-side angle, apex / bevel	20.5–20.8/ 19.5–20.8°
Edge B per-side angle, apex / bevel	20.6–21.6/ 18.7–21.7°
Grind ends from the point, A / B	3.0 / 3.1 mm
Unsharpened plunge, A / B	9.7 / 11.9 mm
Serration depth, A / B	1.63 / 1.41 mm
Scallops, A / B	3 / 5
Weakest tooth SF, A / B (teeth checked)	3.47 / 2.83 (2 / 4)
Solid tip SF (1.82 N·m twist)	0.64
Lateral capacity (search plate mesh)	337 N
Grip SF, worst of four cases (formula)	0.73
Human-max twist SF (search plate mesh)	1.09
Spine 3427 N SF (search plate mesh)	1.44
Ring bearing / back capacity	8,677 / 1,686 N
Fatigue SF / yield-sequence ratio (as coded)	0.94 / 0.56
Buckling factor λ (analytic)	11.2

6.5 Edges and serrations

Fig. 4 plots the blade section at four stations of each edge, measured along the true edge normal with the exact decoder law. Every station of an edge lies on one line: edge A at 20.5–20.8° per side and edge B at 20.6–21.6°, with no slope



Figure 3 | `hz4i013_g00995_d206e3c9`. Top: the knife, and an x-ray with every internal cavity in amber. Bottom: a section across the blade 30 mm from the point, the point, and the spine edge with its serrations. Cycles renders of the exported meshes; the blade is meshed at 0.05 mm from the exact decoder law, the rest at 0.2 mm voxels.

Table 6 | The four masses of the best individual

Definition	Mass (g)
Search voxel count, $h = 0.4$ mm	56.0
Ranking voxel count, $h = 0.2$ mm (scored)	57.0
Print mesh, watertight, 0.2 mm (of record)	59.3
Upload mesh in cad/ (coarser, under 50 MiB)	59.7

steeper than 20.9° and 21.9° within 2 mm of the edge. Both edges are serrated over the rear part of the blade, and the grinds run on through the plain point region to 3.0 and 3.1 mm from the point. Of the teeth between scallops (2 on edge A and 4 on edge B), the weakest has a safety factor of 2.83 under the 180 N snag and 60 N pry, which meets the target of 1; the end tooth of each span, bounded by a scallop on one side only, is not checked.

7 Limits of the result

> **What this result is not**
>
> It is a computational design. Nothing has been printed, heat-treated, sharpened or tested, and no safety factor here is a qualification.

- **Strength against the targets.** Against a target of 1, the solid tip (0.64) is below the target; the grip under the spine load (worst of four cases by the beam formula 0.73; spine load by voxel FE 0.74) is below the target; the human-maximum twist (plate 1.09, voxel FE 1.69) meets the target; fatigue (0.94) is below the target; and the yield-sequence ratio (0.56 against 1.2) is below the target. Lateral capacity meets the target and the ring meets the target.
- **Mass.** The print mesh weighs 59.3 g, which is above the 50 g goal; the score used the 57.0 g ranking-grid count (Table 6).
- **Assumed loads.** The tooth snag and pry loads are

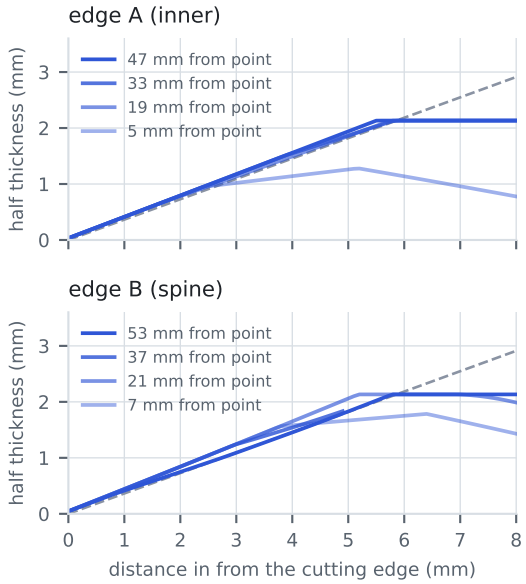


Figure 4 | Upper half of the blade section at four stations of each edge, true scale, against a 20° per-side grind (dashed). Profiles stop where a lightening slot begins.

conservative research assumptions, not measured use loads. The buckling factor is an analytic lateral-torsional estimate [8], not an eigenvalue solve.

- **Models.** Voxel finite elements resolve the printed geometry only to $h = 0.4$ mm and report percentiles, not peaks; the plate and beam surrogates of the search differ from them (Table 2). Surfaces come from marching tetrahedra on the voxel SDF [5], except the blade, which is re-meshed at 0.05 mm from the exact decoder law with marching cubes [6]. Two objective terms are held fixed or capped in this release (Section 3), and the buckling factor is an analytic estimate. The plate capacities do not converge under mesh refinement, and the tip model reads its thickness raster half a cell off (Section 6.4).
- **Coarse search grid.** The archive was selected at $h = 0.4$ mm, which undercounts the mass of some designs (Table 3); only the final ranking is measured at $h = 0.2$ mm, so designs that the coarse grid scored low were never re-checked.
- **Definitions and credit.** The yield-sequence ratio mixes two load cases and the fatigue stress applies the thin-wall knock-down twice (Section 6.4). Tier 1 credits a watertight mesh it does not build; only Tier 2 censuses one. The guard outline reuses tables from the earlier design f075, so that part of the shape is a human prior.
- **One shortened cohort.** The release cohort is 2,500 evaluations per island on one host; another seed or a longer run may find a better design.

8 Reproducibility

The cohort ran at commit `80e669e5b558` with decoder hash `a203278d80d2`; the decoder and g3-evolve are unchanged since. The winner's 179-gene row is evidence/g3/v3.0.0/winner_genome.csv; every island's logged history, the ranking records, the edge profile, the blade facts and the finite-element results sit next to it under evidence/g3/v3.0.0/, listed with their SHA-256 in SHA256SUMS. The receipt release.json is the single source of every computed number in this paper and in the repository README: g3-release-tex turns it into the macros used here, and ops/ci/gen_readme.py into the README. The cohort is reproduced by building g3-evolve from this release and running island $n = 1, \ldots, 24$ with --init 64 --evals 2500, --seed $4000 + n$, --host 4 --island n, --commit `80e669e5b558`, --h 0.4, --tier1-gate 62, --tier2-gate 71 and --log-every 50 (the seed, host and island set the random stream, and the commit enters every genome hash). g3-pilot tier2 at $h = 0.2$ re-ranks the elites, g3-pilot hconv repeats the grid check, and g3-pilot export rebuilds the watertight envelope and print mesh from the genome row. The binary STL files (millimetres) are listed in Table 7.

9 Conclusion

Putting the edge and serration rules into the decoder made every design the search could express a knife with a correctly ground, double-serrated two-sided blade, and the search then traded mass, strength and ergonomics within that space. The search also exposed a gap in the measurement chain: on its coarse grid, a design could look lighter than it would print.

Table 7 | CAD files of the best individual (binary STL, millimetres; each file name starts with the design identifier)

File	Triangles	MiB	Where
`AMbuild_mm.stl`	1,001,372	48	git and release
`envelope_mm.stl`	911,132	43	git and release
`AMbuild_h0.2_mm.stl`	5,373,548	256	release
`envelope_h0.2_mm.stl`	2,063,960	98	release
`envelope_fineblade_mm.stl`	3,843,106	183	release

Re-measuring the ranking on the print grid removed that bias from the final ranking, but not from the search, which still selected on the coarse grid. The best individual of the release, `hz4i013_g00995_d206e3c9`, scores 80.55 with an ergonomic multiplier of 0.988 (scored at its 57.0 g ranking-grid mass), and its print mesh weighs 59.3 g. The limits in Section 7 set the next work: a stronger sharp point, a stiffer hollow grip under the spine load, a search grid fine enough that it cannot be gamed, and a longer cohort, all measured the same way.

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